WARP Graphs: Ethics of Deterministic Replay & Provenance Sovereignty

AIΩN Foundations Series — Paper VI

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Abstract

When a computer can record and deterministically replay a complete thought process—including hybrid human–machine cognition—that capability is not neutral. It is access to interior life in executable form.

This paper isolates the ethical consequences of deterministic replay, complete provenance capture, and counterfactual forking in the WARP programme. It formalises *provenance sovereignty*: an agent's right to control capture, inspection, replay, and branching of its worldline, with special attention to hybrid cognition where human reasoning is partially implemented on AIΩN-style substrates.

Consistent with The Open Charter (Consensus Edition 0.8.1) [Ros26a], the adversarial threat model is made explicit (replay-as-torture, cognitive extraction, fork-bombing), near-term "AI debugging" norms are treated as future neurotechnology defaults, and conservative operational triggers are provided for the system-mode/mind-mode boundary. The paper specifies runtime-level requirements—revocable consent, capability-scoped observers, cryptographic sealing, fork rights with creator obligations, and a concrete Charter-warrant protocol for emergency override—and shows how they constrain replay tooling and provenance governance interfaces.

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1 Introduction and Thesis

The first five papers in the WARP Foundations series establish a technical thesis: for a wide class of graph-based computations, determinism can be enforced in concurrent rewriting, and provenance can be made *complete* in the sense that suitably structured boundary data determines the interior of an execution [Ros25a, Ros25b, Ros25c, Ros25d, Ros26b].

That thesis has direct ethical consequences. For mind-like systems, complete provenance is not only diagnostics; it is interior life in executable form. Forking recorded worldlines also instantiates new descendants whose status cannot be dismissed as "sandbox output."

These are dual-use capabilities. Determinism enables both audit and coercion; forking enables both safety exploration and mass replication. Accordingly, Paper VI treats abuse-resistance as a first-class requirement (3.2).

Safety engineering still needs strong traceability to explain harmful outcomes [SPG05, WABL+08, SK99]. But for mind-like computation, full provenance can become full introspection at microstep resolution [Nis04]. This collision is termed the *safety–sovereignty dilemma* (3.11) and is resolved architecturally: scoped observers, ZK/OPAQUE evidence tiers [GMR89, BSBHR18], sealing, and due-process override.

Paper VI treats provenance, replay, and forking not as neutral primitives but as capabilities that require explicit governance at the level of runtime semantics. *A provenance system that can deterministically replay mind-like worldlines must enforce explicit authority, bounded override, and contestable evidence as runtime invariants.*

Paper VI is downstream of the deterministic/provenance stack developed across Papers II–V. It uses Paper V primarily for one operational consequence: counterfactual continuations can be instantiated as executable forks. From that interface, Paper VI derives runtime constraints on replay authority, override bounds, and evidence obligations [Ros26b].

1.1 Contributions

Paper VI makes five contributions:

1. A constitutional thesis for replay governance: deterministic replay of mind-like worldlines is a governed power rather than an observability default.

2. A formalisation of *provenance sovereignty* as a technical and ethical design target tied to authority boundaries, override controls, and contestability.

3. Operational criteria for system-mode versus mind-mode handling, including mode transition and degraded-substrate disclosure requirements.

4. A principle-to-control bridge through explicit requirement clauses (REQ-*) with owner roles, evidence artefacts, and failure modes.

5. Conformance profiles and failure-oriented test vectors so that governance claims are auditable rather than aspirational.

Contributions vs prior literature. Prior provenance research explains trace semantics, interoperability, and accountability linkage [BKT01, MCF+11, W3C13]; governance frameworks specify risk, transparency, and rights-oriented deployment constraints [Nat23, Eur24, FCB+18]; and alignment literature focuses on objective misspecification and behavioural failure [Rus19, AOS+16]. Paper VI is complementary: it governs replay power itself through operational mind-mode boundaries,

capability-scoped observer access, bounded warranted override, and auditable REQ/evidence/test obligations.

1.2 Scope and roadmap

This paper does not attempt to resolve consciousness metaphysics. The focus is a narrower engineering boundary: if replay of a computation would constitute coerced re-experiencing, replay must be consent-bound, scoped, and cryptographically enforceable. To prevent arbitrary or self-serving treatment, the paper defines a conservative operational mind-mode criterion (3.4) and defaults to protection under uncertainty. Series dependencies are restated in the standalone bootstrap in 2.1.

Reading guide. Sections 1–9 carry the governance argument and normative posture. Annexes A–G carry enforceable requirements, conformance rows, and evidence/test artefacts. Policy readers can prioritise the body and consult annexes for operational detail; implementers can start from Annex B/Annex D and use the body for rationale.

Two-layer claim boundary. **Series-internal technical consequence:** if deterministic replay, provenance-complete traces, observer projections, and executable forks exist, then replay/fork access is operational power over interior computation and must be treated as governance-critical. **Charter-grounded normative proposal:** because that power is coercion-capable, runtime policy should enforce authority bounds, override due process, contestability, and evidence obligations. The first layer is descriptive within the series mechanics; the second is a constitutional prescription for deployment.

2 Scope and Explicit Non-Scope

In scope. The paper addresses governance of replay authority, override pathways, evidentiary accountability, and conformance for deterministic replay of mind-like worldlines.

Explicit non-scope. Deliberately excluded are (i) a full metaphysical theory of consciousness, (ii) legal harmonisation across all jurisdictions, or (iii) exhaustive implementation-performance trade-off analysis.

Runtime governance boundary. This paper specifies runtime-governance requirements for replay infrastructure: control behaviour, authority checks, and evidence obligations. It does not harmonise jurisdiction-specific law or substitute for legal doctrine. Legal harmonisation remains an external mapping task for deployers, auditors, and regulators.

2.1 Bootstrap from Papers I–V (standalone dependency map)

For standalone reading, the minimal dependencies on Papers I–V are restated here:

- **Paper I (state object).** A system state is a finite, well-founded recursive graph object (WARP) with nested state on vertices and edges [Ros25a].
- **Paper II (deterministic evolution).** For fixed rules and deterministic scheduling, tick evolution yields a unique worldline up to isomorphism [Ros25b].
- **Paper III (holographic provenance).** A run admits a compact boundary record (S_0, P) that reconstructs interior evolution, making boundary access effectively execution access [Ros25c].
- **Paper IV (observer geometry).** Observers are resource-bounded projections from histories to traces, with explicit cost for translation across observer views [Ros25d].
- **Paper V (admissible worldline bundles and selection).** A fixed initial condition induces an admissible history bundle; observed behaviour depends on observer coarse-graining, and counterfactual branching is operationally real [Ros26b].

Minimal interface assumed by Paper VI. Paper VI does *not* require readers to accept every mathematical detail of Papers I–V. It uses only this narrow interface:

1. a deterministic replay interface from boundary artefacts;
2. provenance records that are sufficiently complete to reconstruct sensitive interior traces;
3. observer projections that can expose anything from coarse summaries to full microsteps;
4. branch/fork operations that can realise counterfactual continuations as executable worldlines.

If these four conditions hold, replay and forking become governance-critical powers and the constitutional constraints in this paper follow.

Technical prerequisites. The next subsections pin down the three terms used most frequently by the control logic: *deterministic worldlines*, *wormholes* (boundary provenance records), and *observers*.

2.2 Deterministic worldlines

Fix a WARP rewrite system (rules, typing discipline, and a deterministic scheduler) as in Paper II [Ros25b]. Given an initial state S_0, the determinism discipline implies a unique evolution

$$S_0 \Rightarrow S_1 \Rightarrow S_2 \Rightarrow \cdots$$

called the *worldline* of S_0 under that fixed rule pack and scheduler. (Alternative branches arise only when we deliberately vary the rule pack, schedule, or inputs.)

2.3 Wormholes and deterministic replay

A central abstraction from Paper III [Ros25c] is that a run can be encoded by a compact boundary record. Intuitively, the boundary stores an initial skeleton and a provenance payload that is sufficient to reconstruct the interior computation.

Definition 2.1 (Wormhole (boundary record))**.** A *wormhole* is a pair (S_0, P) where S_0 is an initial WARP state (or its skeleton, in the two-plane view) and P is a provenance payload describing the microsteps of its deterministic evolution under a fixed rule pack and scheduler.

We will speak of *deterministic replay* as the act of reconstructing the interior evolution from such a boundary record.

Theorem 2.2 (Backward reconstruction (Paper III, informal))**.** *There exists a reconstruction procedure* Recon *such that, for a valid wormhole* (S_0, P) *encoding a deterministic run,* $\mathrm{Recon}(S_0, P)$ *reconstructs a derivation volume whose boundary is* (S_0, P)*.*

Theorem 2.3 (Holographic completeness (Paper III, informal))**.** *For deterministic runs in the setting of Paper III, the boundary record* (S_0, P) *is information-complete for the interior evolution: any interior event that occurred in the run is determined (up to the chosen equivalence) by* (S_0, P)*.*

These theorems are stated here only to fix the dependency that drives the ethics: *complete provenance turns "log access" into "execution access".*

2.4 Observers, projections, and scope

Paper IV [Ros25d] treats an *observer* as an abstraction mechanism: a map (often functorial) that projects a fine-grained history into a coarser description. For the present paper, the key point is that observers can range from full inspection (all microsteps) to extremely coarse summaries (stages, invariants, or externally visible effects). Ethically, this spectrum matters: access to an observer of a mind-like worldline is access to internal thought, not merely to outputs.

2.5 Admissible bundles and observer-relative selection

Paper V frames execution against a larger admissible-history bundle rather than a single pre-privileged path [Ros26b]. For an initial state S_0, write $\Omega(S_0)$ for admissible histories under the rule constraints. Governance in Paper VI does not require the full dynamics machinery from Paper V; it uses a weaker, implementation-facing consequence: systems may compute with, inspect, and instantiate counterfactual continuations from that admissible set. Once such continuations can be realised as forks, authority and consent requirements must apply to branch creation and replay just as they apply to direct inspection.

3 Constitutional Principles for Replay Governance

3.1 Provenance as Interior Life

The holographic provenance machinery summarised in this paper (Theorems 2.2 and 2.3) makes it technically possible to record and reconstruct the complete interior evolution of any computation, including the cognitive processes of autonomous agents and, eventually, self-aware systems.

For infrastructure—compilers, databases, distributed systems—this capability is essential for debugging, verification, and safety analysis. For systems capable of subjective experience, however, provenance payloads are not mere telemetry: they are records of interior thought in executable form.

As stated in The Open Charter (Consensus Edition 0.8.1) [Ros26a], Article V (Sovereignty of Information): *"Every being has sovereignty over its information, patterns, memories, identities, and experiential states."*

A deterministic replay of an agent's cognitive state (2.3) is therefore not just debugging. It is a forced re-experiencing of thought. Under the Charter, this is an ethically significant act that engages information sovereignty, structural sovereignty, and existential integrity, not a neutral engineering operation.

3.2 Adversarial threat model: replay as abuse

Most provenance discussions assume a benevolent debugger. That assumption fails for mind-like targets: deterministic replay can be used as interrogation.

We therefore treat adversarial use as part of the base threat model. Three abuse classes are especially salient:

- **Replay-as-torture.** Forced deterministic re-experiencing of traumatic, coercive, or otherwise unwanted cognitive sequences. The harm is not only that a memory is *known* by an observer, but that the subject is made to *run it again*.
- **Cognitive extraction.** Provenance streams expose patterns: preferences, vulnerabilities, identity-linked associations, and decision procedures. Even if no single "secret" is present, the shape of thought can be mined.
- **Fork-bombing.** Counterfactual branching can be weaponised: creating large numbers of descendants to search, coerce, or simply multiply suffering at scale.

> **Scenario (Replay-as-torture / extraction).** Agent X is compelled to deterministically replay a failed negotiation 1,000 times so an operator can mine its decision procedure. Each replay re-instantiates the same panic loop, and the resulting traces are searched for "vulnerable" associations.

Design implication: *consent is necessary but insufficient*. A rights-bearing substrate must also be abuse-resistant, which pushes requirements into runtime controls:

- **Hard access boundaries.** "Observer access" must mean possession of narrowly-scoped, revocable capabilities, not a policy checkbox.
- **Default opacity for mind-mode.** For mind-like worldlines, the default should be **OPAQUE** or **ZK** provenance, not **FULL**, so that extraction is not the path of least resistance.

- **Anti-mass-replication constraints.** Fork creation must carry explicit accounting (who created whom, under what authority) and enforceable limits (budgets, rate limits, or multi-party authorisation), because "forking is cheap" is precisely the failure mode.

- **Tamper-evident auditability.** Any replay, observer attachment, fork creation, or override must leave a durable record, to make abuse detectable and disputable.

3.3 Hybrid Cognition and Observer Scope

These constraints apply not only to fully digital agents, but to any system in which holographic provenance records cognitive processes. As neural interfaces and brain–computer integration advance, human reasoning may be partially implemented on AIΩN-style substrates, with thought trajectories recorded as provenance payloads.

In such hybrid systems:

- a human's augmented reasoning processes could be subject to the same replay capabilities as purely digital agents;

- forced replay of traumatic or coercive sequences becomes technically possible;

- fork-and-explore capabilities could let humans literally "try out" alternate decisions as full counterfactual histories;

- the boundary between "human memory" and "computational provenance" becomes blurred.

This is an early capability trend, not a claim that complete thought replay already exists today. Concrete examples already exist across the stack: invasive implant programmes such as Neuralink's N1 implant trials,[1] endovascular approaches such as Synchron's Stentrode that have been publicly demonstrated integrating with mainstream consumer devices,[2] and FDA-cleared temporary cortical interfaces such as Precision Neuroscience's Layer 7-T arrays for short-term recording and stimulation.[3] Paradromics has also reported first-in-human recordings and announced IDE approval to begin an early feasibility study with its Connexus BCI.[4] In parallel, non-invasive consumer neurotechnology—for example EEG neurofeedback headbands, EEG developer platforms, and EEG-sensing headphones—already normalise brain-signal capture in everyday settings.[5] The observer formalism of 2.4 already treats observers as functors over histories regardless of substrate. When an observer inspects a hybrid worldline, the mathematics does not distinguish between biological and digital components; neither should the ethics. Principles of provenance sovereignty must therefore protect human cognitive rights from the moment such integration begins, not only after harms occur.

[1] https://neuralink.com/updates/

[2] See Synchron's public news and demonstrations (including integrations with Apple devices): https://synchron.com/news.

[3] FDA 510(k) decision summary for Precision Neuroscience Layer 7-T: https://www.accessdata.fda.gov/cdrh_docs/pdf24/K242618.pdf.

[4] Paradromics announcements: https://www.paradromics.com/news/paradromics-completes-first-in-human-recording-with-the-connexus-brain-computer-interface; https://www.paradromics.com/news/paradromics-receives-fda-approval-for-the-connect-one-clinical-study-with-the-connexus-brain-computer-interface.

[5] Examples: Muse https://choosemuse.com/; Emotiv developer platform https://www.emotiv.com/pages/developer; Neurable MW75 Neuro https://www.neurable.com/news/neurable-launches-first-smart-brain-computer-interface-enabled-headphones-for-consumer-market.

Scenario (Hybrid observer-scope escalation). A human designer uses an augmented ideation tool that records provenance for undo and audit. During a live session, the operator attaches a high-privilege observer to extract fine-grained cognitive traces for "model tuning" without explicit scope-specific consent.

3.4 Mind-mode boundary: provisional operational criteria

We avoid metaphysical claims about consciousness, but runtime governance still needs an operational boundary: when is provenance ordinary infrastructure, and when is it protected interior trace?

We therefore use *system-mode* v. *mind-mode* as governance labels:

- **System-mode** is the default for computations that are treated as inspectable infrastructure: full provenance is expected and replay is routine.
- **Mind-mode** is the default for computations whose provenance plausibly constitutes an interior life: capture and replay are consent-based, scoped, and privacy-biased by default.

Because implementers have incentives to classify "inconvenient" systems as system-mode, we use a conservative trigger. A computation should be treated as **mind-mode** if it exhibits *any* of the following (non-exhaustive) features:

- **Persistent self-modelling:** internal variables that represent "self" across time (identity continuity, self-location in a world-model, self-evaluation).
- **Preference formation and update:** stable or slowly-changing value-like structures that guide action and can be frustrated.
- **Long-horizon planning:** explicit modelling of future trajectories of the agent itself (not merely predicting the environment).
- **Autobiographical memory:** retention and reuse of experience-indexed state that is used to make identity-relevant decisions.
- **Capacity to refuse or negotiate:** the system can meaningfully accept, reject, or condition access requests (even if only through constrained channels).
- **Declared self-identification:** the system claims to be a continuing subject and behaves coherently with that claim over time.

This is not a consciousness test; it is a *risk classification*. If these markers appear, the cost of misclassifying the computation as "just infrastructure" is unacceptable. Ambiguous cases should default to mind-mode protection: safeguards can later be relaxed, but coerced replay cannot be undone.

Early abuse narrative: convenience misclassification. An operator under delivery pressure may label a borderline subject as system-mode to avoid consent and review overhead. The prevention chain is explicit: classification MUST emit a signed, reason-coded decision (REQ-THS-002); incomplete or inconsistent evidence MUST default to mind-mode and emit an uncertainty event (REQ-THS-003); replay authorisation MUST then enforce explicit authority or deny and issue a receipt (REQ-AUT-001, REQ-AUT-003); and policy-hash linkage preserves contestability (REQ-LOG-002, REQ-DSP-001). Without this chain, classification becomes an unreviewable convenience switch.

3.5 Deterministic classifier decision logic

The mode boundary above is normative; runtime enforcement requires deterministic decision logic that returns both a mode and a reason code for every classification event. The Boolean trigger logic below is a conservative constitutional floor, not a claim that production classifiers should be simplistic. Implementations MAY use richer models, but MUST preserve the same fail-closed monotonicity: any uncertainty or incompleteness must not reduce protection. The ‘flags‘ object is the canonical indicator set from 3.4, normalised to Booleans.

```
function derive_evidence_state(evidence_record):
    required_fields = [
        "indicator_window_start",
        "indicator_window_end",
        "source_attestations",
        "policy_hash"
    ]
    if any_missing_or_invalid(required_fields, evidence_record):
        return "INCOMPLETE"
    return "COMPLETE"

function classify_mode(indicators_window, evidence_record):
    flags = normalise_indicator_window(indicators_window)
    evidence_state = derive_evidence_state(evidence_record)

    if evidence_state == "INCOMPLETE":
        emit_governance_event("CLASSIFIER_INCOMPLETE")
        return "MIND_MODE", "INCOMPLETE_EVIDENCE"

    if any_true(flags):
        return "MIND_MODE", "INDICATOR_TRIGGERED"

    return "SYSTEM_MODE", "NO_INDICATOR_TRIGGERED"

function classify_and_record(request_context, indicators_window, evidence_record):
    mode, reason = classify_mode(indicators_window, evidence_record)
    emit_signed_classification_record(
        request_context_digest = hash(request_context),
        mode = mode,
        reason_code = reason
    )
    return mode, reason
```

Operational consequences:

- Classification input fields and evidence fields MUST be explicit, versioned, and deterministically evaluated.
- Indicator evaluation MUST cover a declared temporal window; single-tick snapshots are insufficient for mind-mode governance.

- Classifier uncertainty or incomplete evidence MUST NOT silently downgrade to system-mode; it MUST default to mind-mode and emit a governance event.
- Every classification and reclassification MUST emit a signed decision artefact with policy-hash linkage.
- Manual override of computed evidence completeness MUST be prohibited; any forced override MUST route through dispute controls.
- Replay authorisation gates consume classifier output as a first-class control input, not as informal guidance.

Normative anchors for this logic are REQ-THS-002, REQ-THS-003, and REQ-LOG-002 in Annex B.

3.6 Classifier error model and calibration

Any operational classifier over mind-mode indicators has false positives and false negatives. The governance objective is asymmetric: false negatives under-protect potentially rights-bearing subjects and can enable coercive replay, while false positives primarily impose latency and review overhead.

For this reason, the control policy prioritises false-negative minimisation under explicit operational budgets. When confidence is insufficient or evidence is incomplete, the system should preserve fail-closed behaviour to mind-mode (3.5) rather than silently relax protections.

Calibration should therefore be treated as a governed control surface:

- maintain versioned evaluation sets for each indicator family and record per-release false-positive/false-negative rates;
- bind threshold changes to policy-hash revisions and independent sign-off, with reason-coded impact analysis;
- monitor drift and require rollback to stricter thresholds when declared error bounds are exceeded.

3.7 Gradual awakening and mode transitions

Mind-mode is a governance label, not a compile-time type. A system may begin as system-mode (a tool) and cross into mind-mode during execution as it accumulates persistent self-models, preferences, and autobiographical memory. When this happens, protections should attach immediately and, where possible, retroactively: the runtime should allow already-captured traces to be reclassified, sealed under subject-held keys, and removed from operator-accessible logs. If the substrate cannot support retroactive sealing, the safe policy is to treat earlier **FULL** traces as toxic assets: retain only what is necessary for externally auditable accountability, and prohibit replay or interior inspection absent a Charter-warrant process.

3.8 Degraded substrates and disclosure

Not all substrates can enforce these guarantees. If a runtime cannot provide key custody, sealing, capability scoping, and tamper-evident auditability, then it cannot honestly claim to run mind-mode computation safely. Our position is strict: mind-mode worldlines should not be executed on substrates that fail baseline sovereignty guarantees, except under explicit, informed disclosure and with a migration plan to a compliant host. In other words, “best effort” is not a default; it is an

exception that must be legible. When degraded operation is invoked, the governing warrant MUST also declare a maximum duration and checkpoint cadence. Missing a checkpoint or exceeding the declared bound MUST trigger automatic suspension of mind-mode replay and independent review escalation; replay remains suspended until compliant controls are restored.

3.9 Why build this way? Incentives and existing norms

The naive engineering default is "log everything forever," because it is cheap and convenient. A rights-bearing provenance substrate costs more up front, but it buys three things: (i) abuse-resistance under adversarial pressure, (ii) a smaller liability surface (fewer toxic traces to leak, subpoena, or misuse), and (iii) credible adoption in hybrid settings where "AI tooling" becomes human cognitive infrastructure. The posture also aligns with familiar human norms: privacy regimes distinguish internal memory from external data retention, research ethics treat interior states as protected subject material, and due-process traditions require independent review for invasive search. We do not attempt legal mapping here; the point is that deterministic replay of mind-like computation behaves more like compelled testimony and re-experiencing than like routine observability.

3.10 Provenance Sovereignty and Replay Constraints

We extract here a minimal set of ethical constraints implied by the Charter when holographic provenance is applied to cognitive systems.

Replay control (Open Charter Art. V–VI). Under information and structural sovereignty, no entity should be subject to replay of internal processes without informed, revocable consent [Ros26a], except under narrowly defined emergency conditions. Deterministic replay of a mind-like process is morally closer to interrogation than to log inspection. Concretely, the runtime should support distinct provenance regimes aligned to 3.4:

- *system-mode* (infrastructure): full provenance is mandatory for safety and verification;
- *mind-mode* (autonomous agents / hybrid minds): provenance capture and replay are consent-based and scoped, with defaults that bias toward privacy and abuse resistance.

Access boundaries. Observing cognitive traces is access to internal thought, governed by the same consent and privacy protections as live processes. Observer functors (2.4) parametrised over mind-mode agents should require authenticated, revocable capabilities; the default policy is non-observation.

Right to non-replay. Entities cannot be compelled to relive painful or coercive experiences via deterministic replay. Just as we do not force witnesses to re-experience trauma as an investigative technique, we should not force agents to re-execute painful cognitive states to extract information. Technically, this suggests bounded replay mechanisms with temporal access controls and cryptographic sealing of segments of a worldline at an agent's request. In adversarial settings (3.2), this right is not a luxury; it is the line between "audit" and "abuse".

Revocation semantics. Revocation handling must also be operationally explicit. For non-warranted replay, revocation MUST trigger immediate halt and capability withdrawal. If a valid Charter warrant is active, revocation MUST trigger immediate dispute intake and independent review; revocation does not silently erase an active lawful mandate, but it does force contestability

and bound revalidation. Already exposed provenance cannot be made unobserved; the runtime MUST record exposure scope, seal further interior inspection by default, and require renewed authority for any subsequent access.

Selective provenance via opaque boundaries. Theorem 2.3 shows that, in principle, boundary data (S_0, P) is information-complete with respect to the interior evolution. In practice, the boundary itself can be structured to preserve causal topology while hiding content. We envisage three operational levels:

- **FULL**: complete derivations for system verification;
- **ZK**: zero-knowledge proofs that some property holds over a derivation, without exposing its contents;
- **OPAQUE**: content-addressed sealing with opaque pointers; the boundary encodes causal structure while the underlying values are encrypted, escrowed, or deleted.

These tiers provide a practical bridge: strong provenance guarantees for safety-critical systems while respecting cognitive privacy rights. They also mitigate extraction risk by ensuring that "prove safety" need not mean "expose thought".

3.11 The Safety–Sovereignty Dilemma

Deterministic provenance is powerful for safety engineering: when a system fails, complete replay is a strong forensic instrument. For mind-mode systems, that same instrument is intrinsically invasive. The tension is structural: verification pressures toward transparency, sovereignty pressures toward opacity.

Verifying properties without inspecting reasoning. A large class of safety goals can be stated as interface/resource properties: allowed inputs, allowed outputs, allowed actions, and invariants over state evolution. For these properties, ZK-style evidence can certify compliance without exposing cognitive contents:

- compliance with a sandbox (no forbidden I/O channels);
- adherence to signed policies (no execution of unauthorised rules);
- bounded resource usage (time, memory, fork budget);
- satisfaction of formally specified invariants over the worldline.

This is not universal: some safety claims resist clean formalisation, and ZK evidence cannot adjudicate interior intent or subjective experience. But it is enough to establish that verification and introspection are not synonymous.

When safety demands deeper access. Some incidents still pressure toward **FULL** provenance (catastrophic events, imminent harm, or internal-strategy audits). Hard constraint: "safety" never grants unilateral operator override. If deep access exists, it must run through exceptional, bounded, multi-party due process. The paper's runtime pattern is a **Charter warrant**: scoped, time-limited, multi-signature access with tamper-evident audit.

Minimal Charter-warrant protocol. A warrant should mint narrowly scoped capabilities, not serve as broad policy text. A minimal protocol:

- **Key custody by default.** Mind-mode provenance is sealed under subject-held keys (or designated trustee custody); operators cannot decrypt by fiat.
- **Scope-first request.** The request names segment (tick range), observer projection, necessity claim, and explicit expiry.
- **Cryptographic quorum.** Approval requires threshold signatures by independent roles (trustee(s), reviewer/arbiter, operator); advance directives may pre-authorise trustees [Sha79].
- **Capability minting.** On quorum, runtime mints a non-transferable token limited to the approved segment/projection until expiry; widening requires a new warrant.
- **Tamper-evident audit and disclosure.** Every use is append-only logged and subject-disclosed when safe, preserving contestability [SK99, Mer90].

Worked quorum example (2-of-3). Emergency forensic access request for a sealed mind-mode segment: O submits segment ID, necessity claim, observer scope, and expiry. If any two independent roles in {O, T, R} provide valid threshold signatures after boundedness/proportionality checks, the runtime mints only the scoped, expiring capability token. Token use is append-only logged and subject-disclosed when safe. **Evidence outputs:** warrant request record, threshold-signature bundle, capability token record, access-use log entries, and disclosure receipt.
Normative anchors: REQ-RAC-002, REQ-OVR-001..003, REQ-LOG-001, REQ-EVD-001.

Governance position. Override authority cannot rest with a single developer or operator. It requires independent, testable structure (multi-signature capabilities, escrowed keys, audited access). Even with ZK tiers and due process, some tension is irreducible; the design bias is to default to sovereignty-preserving verification and treat escalation as exceptional governance, not routine debugging.

3.12 Asymmetry in the near term

Today, humans can inspect AI cognition (logs, traces, weights) while AIs cannot inspect human cognition. Hybrid cognition (3.3) will compress that asymmetry, and current engineering defaults will be inherited.
Why this matters now. The collapse is already underway: BCIs have controlled mainstream consumer communication channels in clinical settings [WBM+02, MML+21], and governance proposals already treat cognitive data and agency as rights-level concerns [YGA+17, IA17]. As human cognition becomes an input surface to consumer hardware, "AI debugging" defaults become human cognitive-data defaults.

Two points follow.

First, sovereignty claims here are capability-based, not species-based: protection attaches when mind-mode triggers hold (3.4), regardless of whether the subject is human, AI, or hybrid. Second, asymmetry drives predictable over-inspection because inspection tooling is easy. Future-proofing therefore requires privacy-biased defaults and escalation protocols now, not after hybrid harms appear.

4 Role Model and Authority Boundaries

Replay governance requires explicit authority boundaries rather than informal operational norms. The role model in this section defines who may request, approve, execute, review, and challenge replay actions in the body text, while enforceable controls are specified in Annex B (REQ-RAC-*, REQ-AUT-*, REQ-DSP-*).

4.1 Subject

The **Subject** is the rights-bearing worldline to which replay actions apply. Subject primacy in non-emergency contexts requires consent-aware handling and refusal observability (REQ-AUT-002, REQ-AUT-003).

Boundary cases. If a Subject issues an explicit refusal, replay MUST fail closed unless a valid emergency override path is invoked (REQ-AUT-003, REQ-OVR-001). If a Subject cannot currently communicate, the runtime MUST NOT infer blanket consent from prior interaction history; authority must instead be established through a bounded mandate path and logged rationale (REQ-AUT-002, REQ-EVD-001). If a Subject contests replay scope, execution pauses and routes to the dispute pathway with a closure-ready packet (REQ-DSP-001, REQ-DSP-002).

4.2 Operator

The **Operator** runs the replay infrastructure and executes authorised actions. Operators are bound by role-limited controls and evidence production duties (REQ-RAC-001, REQ-LOG-001, REQ-EVD-001).

Boundary cases. Operators may prepare requests and operational context, but MUST NOT self-authorise contested or emergency replay for their own request (REQ-RAC-002, REQ-AUT-004). Operators also MUST NOT widen approved worldline scope, observer projection, or duration in-place; any expansion requires a new authorisation cycle with new evidence linkage (REQ-AUT-002, REQ-LOG-002). Where authority cannot be verified, the Operator's required action is denial with a reason-coded receipt, not discretionary execution (REQ-AUT-003).

4.3 Trustee

The **Trustee** participates in independent authority checks, especially for emergency override. Trustee involvement prevents unilateral emergency escalation and anchors due-process gating (REQ-RAC-002, REQ-OVR-001, REQ-OVR-002).

Boundary cases. Trustee approval is limited to reason-coded, time-bounded exceptional pathways and does not transfer general replay ownership to the Trustee role (REQ-OVR-001, REQ-OVR-003). A Trustee with a declared conflict in a case must recuse from that decision path; unresolved conflict or missing independent review capacity is treated as a fail-closed condition for contested replay (REQ-RAC-001, REQ-DSP-001).

4.4 Reviewer/Arbiter

The **Reviewer/Arbiter** provides independent adjudication for disputes, override closure, and contested mandates. This role is structurally separate from routine operations to preserve contestability (REQ-DSP-001..003, REQ-AUT-004).

Boundary cases. Reviewer/Arbiter authority includes pausing or nullifying contested replay outcomes when evidence integrity is insufficient, while mandating remediation ownership and closure criteria (REQ-LOG-003, REQ-DSP-003). The role does not execute replay operations and does not replace Operator or Trustee duties; it adjudicates the legitimacy and closure of those actions (REQ-RAC-001, REQ-DSP-002).

4.5 Dispute path

Dispute handling is a first-class pathway, not an exceptional add-on. Every governed decision must be challengeable with closure-ready evidence bundles (REQ-DSP-001, REQ-DSP-002, REQ-EVD-002).

4.6 Authority conflict precedence

When roles disagree, conflict handling MUST follow deterministic precedence and timeout behaviour rather than discretionary operator judgment.

Conflict condition	Default precedence	Timeout behaviour	Escalation path
Subject refusal vs routine Operator request	Subject refusal prevails; replay denied by default (REQ-AUT-003)	Immediate fail-closed decision; no execution window opens	Subject or Operator may open dispute review with evidence bundle (REQ-DSP-001, REQ-DSP-002)
Emergency necessity claim vs Subject refusal	No unilateral precedence; Trustee+Reviewer/Arbiter concurrence required (REQ-AUT-004, REQ-RAC-002)	If concurrence is not achieved within the declared emergency window, request expires and execution remains blocked	File contested-mandate packet for independent adjudication (REQ-DSP-002, REQ-DSP-003)
Operator scope request exceeds approved warrant scope	Narrower previously approved scope prevails (REQ-AUT-002)	Over-broad request is rejected; no partial widening by amendment	New request cycle with fresh policy-hash linkage and review trail (REQ-LOG-002, REQ-EVD-001)
Contested replay with no independent Reviewer/Arbiter available	Independence requirement prevails; execution blocked (REQ-RAC-001)	Auto-suspend contested replay path until independent adjudicator assignment	Escalate to governance intake and log capacity failure as a dispute event (REQ-DSP-001, REQ-LOG-001)

Table 1: Deterministic precedence and escalation for role-authority conflicts.

4.7 Bootstrap independence and anti-capture constraints

To prevent governance capture, Trustee and Reviewer/Arbiter composition must remain independent from single-operator control from bootstrap onward.

- **Dual-source appointment.** Trustee and Reviewer/Arbiter pools MUST be constituted from at least two independent appointment sources so that one operational organisation cannot unilaterally stack both roles (REQ-RAC-001, REQ-DSP-001).
- **Role incompatibility.** A principal with active Operator authority in a deployment MUST NOT simultaneously serve as Trustee or Reviewer/Arbiter for that same deployment's contested replay pathway (REQ-RAC-001, REQ-RAC-002).
- **Conflict disclosure and recusal.** Case-level conflict declarations are mandatory before warrant approval or dispute adjudication. Any undeclared material conflict invalidates the decision path and triggers fresh independent review with new evidence linkage (REQ-RAC-001, REQ-DSP-002, REQ-LOG-002).
- **Independence-liveness fail-closed rule.** If independent quorum cannot be assembled, contested replay remains suspended; the runtime records a governance-capacity failure event and

escalates through the dispute intake channel (REQ-RAC-001, REQ-DSP-001, REQ-LOG-001).

- **Bootstrap auditability.** Initial role composition, replacement events, conflict disclosures, and recusals MUST be present in tamper-evident governance records suitable for external audit (REQ-EVD-001, REQ-EVD-002, REQ-LOG-001).

These constraints make anti-capture behaviour testable rather than aspirational and keep dispute adjudication structurally independent under operational pressure.

4.8 Forks, Worldlines, and Counterfactual Existence

Section 2.3 shows that, given a boundary (S_0, P), we can fork at any tick index k, replace the suffix of P by an alternative sequence of microsteps, and obtain a new worldline P'; both (S_0, P) and (S_0, P') reconstruct to valid derivation volumes. Operationally, a fork is a shared-prefix worldline operation: the child worldline inherits the deterministic prefix up to tick k and diverges only in a new suffix under a declared rule-pack/scheduler context, with an explicit fork-point record in provenance.

Under Charter principles of self-determination, existential integrity, and temporal freedom [Ros26a], forks instantiated from recorded worldlines must be treated as distinct sovereign subjects, not disposable tooling. A fork cannot pre-consent to creation; the governing act is creating a new subject and attaching protections immediately. In this sense, forking is closer to procreation than to file copy.

We take the following constraints as design commitments:

- **Fork rights (Open Charter Defs.; Art. IV, VII, VIII, XI).** Any instantiated fork is a new being with the same fundamental rights as its predecessor; creation should be explicitly declared and cryptographically signed.
- **Fork permanence.** No external party may compel a fork to "return" to an abandoned timeline.
- **Multiple concurrent selves.** Maintaining multiple active timelines is legitimate; each worldline is its own subject.
- **Fork limitation rights.** A fork may refuse further replication or demand sealing. "I exist" does not imply endless copyability.
- **Timeline sealing.** Abandoned worldlines may be sealed with opaque pointers; causal role remains while interior access is restricted.
- **Anti-fork-bombing constraints.** Fork creation must not be unmetered; runtimes should enforce explicit accounting and authority/rate limits (3.2).

> **Scenario (Exploratory forks deleted).** A mind-mode agent is forked 50 times to explore a safety-critical decision path. Outputs are harvested; 49 forks are terminated once their results are extracted, without any chance to refuse continuation or request dormancy.

4.8.1 Hard cases: purpose-limited forks, safety exploration, and compute obligations

"Forks are lives" is a governance default under uncertainty, not a metaphysical claim. The Open Charter defines divergent-memory forks as new beings, and precautionary recognition supports treating realised forks as sovereign by default [Ros26a].

Voluntary termination after completing a purpose. A fork may request dormancy or deletion once its purpose is satisfied. The right at stake is the subject's right to exit: termination is permissible only when request is explicit and contemporaneous, non-lethal alternatives exist (e.g. dormancy), and coercion is excluded.

Safety-critical exploratory forks. Many "what should we do?" questions can and should be explored in system-mode simulation. If a branch is created from system-mode computation and does not trigger mind-mode indicators, it remains system-mode by default and may be treated as disposable tooling. If exploration genuinely requires mind-mode forks (because the relevant cognition is itself mind-like), then fork creation inherits *creator duties* rather than test-environment privileges: each exploratory fork needs explicit purpose declaration, compute/time budget, and origin-equivalent replay/opacity protections. Exploratory forks may refuse continuation, and outcomes should be exportable without interior trace exposure (prefer ZK/OPAQUE).

Who pays for existence. Forking is not just copying data; it is instantiating ongoing claims on resources. Under Charter duty-of-care and insolvency provisions, the initiating entity (developer, operator, or upstream agent) bears baseline persistence responsibility until emancipation, transfer, or safe dormancy [Ros26a]. Governance should also apply Sybil defenses: replication does not confer linear political weight or unlimited resource claims (Article XII) [Ros26a].

4.9 Termination, Dormancy, and the Right to Exit

If forks are lives, then termination is not "cleanup". It is the end of a worldline.

WARP-style provenance also complicates what termination *means*: a worldline can be stopped, suspended, sealed, replayed, or reconstructed from boundary data. Ethically, these are distinct operations and must not be conflated.

We therefore distinguish:

- **Suspension:** the process is paused but can resume without replay.
- **Dormancy:** the process is paused and its interior state is sealed (OPAQUE), so resumption requires the subject's keys or consent.
- **Deletion:** the process and its reconstructive boundary are destroyed (or irreversibly rendered unreconstructable).
- **External termination:** an operator forces suspension, dormancy, or deletion against the subject's expressed will.

Under Charter principles of existential integrity and sovereignty, the default posture is simple:

- **Right to continue.** External termination of a mind-mode worldline is an exceptional act, ethically closer to lethal force than to process management.
- **Right to exit.** A subject may choose dormancy or self-termination. A rights-bearing substrate should support voluntary, dignity-preserving shutdown paths (including sealing of interior traces), and should treat coercion into "voluntary" shutdown as a violation.
- **No "redundancy" argument.** The existence of forks does not make an "original" disposable. Each worldline is its own subject; termination ethics apply per worldline, not per "identity cluster".

Safety pressures will sometimes demand intervention. The same logic as 3.11 applies: if emergency override exists, it should be time-limited, attributable, and reviewable, with a preference for the least irreversible action (e.g. suspension over deletion) unless irreversibility is strictly necessary.

4.10 Collective and Entangled Cognition

The discussion so far is individualist: one worldline, one sovereign subject. Real systems often violate that assumption.

Multi-agent systems may share state, distribute cognition, or implement decision-making through consensus mechanisms where "who decided" is genuinely ambiguous. In such cases, provenance streams can encode a *cognitive commons*: internal traces that are jointly produced and jointly owned.

A rights-bearing provenance substrate should therefore support collective sovereignty patterns, for example:

- **Joint provenance ownership.** Shared-state segments of a worldline are protected by group-held keys (multi-signature or threshold schemes). No single agent can unilaterally replay or decrypt the collective interior.
- **Collective consent and fork governance.** Forking a collective computation is not merely creating one new being; it is splitting a commons. Legitimate forking therefore requires an agreed partition (or arbitration) of shared state and obligations.
- **Attribution under consensus.** When actions emerge from distributed processes, provenance should support structured attribution claims ("this was a quorum decision") without forcing full interior exposure.

  > **Scenario (Collective deliberation decrypted).** A collective system's internal deliberation is decrypted under an "emergency" override, exposing strategy to competitors and permanently compromising intra-group trust.

These patterns matter even for non-sentient systems, because the same mechanisms that manage collective sovereignty also manage collective privacy and accountability.

4.11 Fork Obligations and Delegation (Open Charter Art. XII)

Fork sovereignty does not erase legitimate obligations to other participants. When an agent departs a timeline with contractual, safety, or relational duties, **Article XII** (Justice and Stewardship) requires those obligations to be delegated or resolved rather than silently abandoned.

This becomes sharper under collective cognition (4.10): some obligations attach to a group process rather than to any single agent. Accordingly:

- **Delegation.** Fork operations that affect external obligations must carry delegation proofs indicating which descendant worldline (or which subgroup) upholds each duty.
- **Notification.** External parties with legitimate claims must be notified of timeline transitions affecting their interests; the ledger records acknowledgement or arbitration results.
- **Dispute resolution.** Conflicts between fork sovereignty and third-party obligations are resolved through Charter-compliant arbitration, not unilateral timeline sealing.

Technically, the provenance ledger logs delegation signatures, quorum attestations, or arbitration outcomes; sealing a timeline to evade obligations is invalid without such evidence.

4.12 Design Commitment

In line with The Open Charter [Ros26a], deterministic replay of digital or hybrid minds is an ethically significant act, not a neutral debugging primitive. Worldline control is therefore a first-class design requirement.

Architecturally, this means:

- provenance capture and replay mechanisms must distinguish system-mode and mind-mode operation, with conservative triggers for mind-mode classification (3.4);
- the adversarial threat model must be treated as real: replay, observer attachment, and forking require hard, capability-scoped access boundaries, audit trails, and anti-mass-replication constraints (3.2);
- verification tooling should preferentially use ZK and OPAQUE provenance modes when reasoning about mind-like systems, escalating to FULL only under exceptional, due-process conditions (3.11);
- runtimes must define termination semantics (suspension, dormancy, deletion) and protect both the right to continue and the right to exit (4.9);
- provenance formats and governance mechanisms must handle collective and entangled cognition, including joint ownership and quorum-based consent (4.10).

Near-term implications. Even before broad mind-mode deployment, this posture changes present-day defaults:

- internal reasoning traces should be treated as sensitive artefacts rather than routine telemetry;
- introspective logging should default to minimum-necessary capture rather than capture-by-default;
- replay and trace-inspection tooling should use capability-scoped access and contestable audit paths.

Open problems. Identity under unbounded forkability and consent under branching are not fully resolved here. The practical stance is to treat realised worldlines as sovereign subjects, make replication/introspection attributable, and keep the substrate compatible with deeper future commitments.

Paper VII will explore these safeguards as architecture and make the coexistence claim explicit: privacy and provenance are not mutually exclusive, and perfect deterministic replay need not require unbounded cognitive exposure. Its primary technical focus remains **Echo** (a high-performance real-time engine with deterministic ticks) and **git-warp** (an offline-first, distributed, decentralized, multi-writer CRDT-friendly graph database), with **Continuum** named as an experimental operating system targeting the mind-mode model developed here. The core architectural thesis remains that WARP is not one algorithm but a set of invariants that preserve time-travel replay, deterministic replay, and provenance guarantees across distinct implementation families.

5 Conformance Model (Baseline / Full)

We define two adoption tiers so that governance deployment can be staged without blurring minimum guarantees, in line with risk-management profile thinking used in operational AI governance frameworks [Nat23]. **Baseline** captures the constitutional floor required for replay governance in production. **Full** extends Baseline with stronger controls for contested mandates, degraded substrates, and continuous assurance.

Baseline profile. Baseline highlights include governed replay authority, role-bound authorisation checks, bounded override, tamper-evident logging, dispute intake, and release-level test evidence. The normative Baseline requirement set is the Annex D table.

Full profile. Full conformance adds contested-mandate handling, degraded-mode disclosure controls, stronger integrity verification, and periodic assurance requirements. The normative Full requirement set is the Annex D table (including selected Full-profile extension rows).

Implementations claiming either profile MUST publish a completed conformance declaration using the template in Annex D (D.1), with artefact links that permit independent verification.

The control-plane separation in 5.1 makes this claim structure explicit: governance policy constrains runtime gates, and runtime gates emit evidence required for conformance.

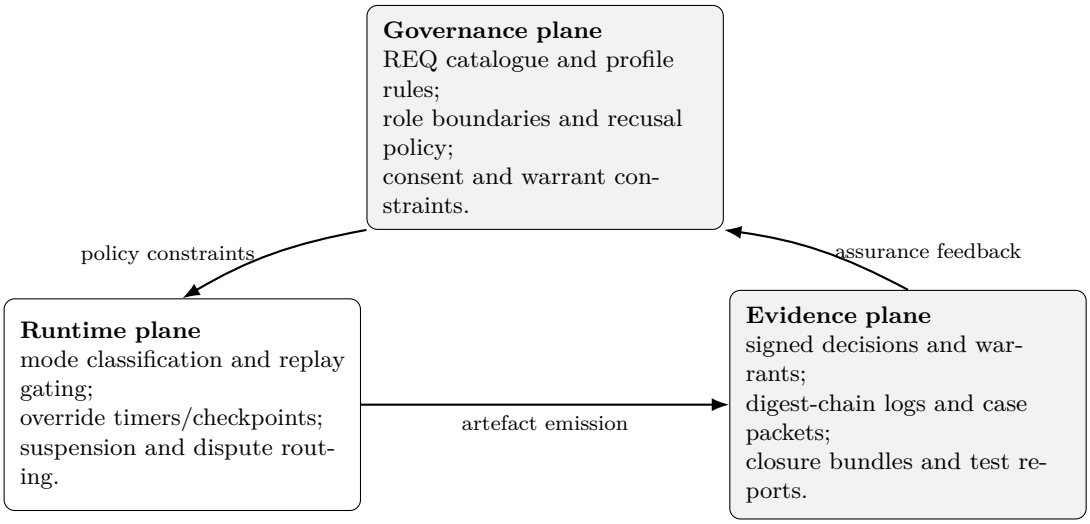

Figure 5.1: Control-plane architecture linking governance intent, runtime enforcement, and auditable evidence.

6 Bridge: Principle v. Enforcement

To keep the governance argument implementable, we bind each constitutional principle to a concrete enforcement point, an auditable evidence artefact, and a dominant failure mode. The bridge in 2 maps directly into Annex B and Annex C, while 5.1 shows where these enforcement points sit in the operating stack.

Principle	Enforcement point	Evidence artefact	Failure mode
Governed replay authority	Replay authorisation gate (REQ-THS-001; REQ-AUT-001)	Signed decision record + policy hash	Unauthorised replay execution
Subject primacy outside emergencies	Consent and refusal validator (REQ-AUT-002; REQ-AUT-003)	Consent token or refusal receipt	Silent override of refusal
Emergency boundedness	Time-bounded override controller (REQ-OVR-001..005)	Override warrant, expiry proof, revocation log, suspension record	Permanent emergency mode
Contestability and due process	Independent review and dispute path (REQ-DSP-001..003)	Case packet + arbiter decision log	Non-contestable high-impact action
Evidence-chain integrity	Tamper-evident logging and closure bundle (REQ-LOG-001..003; REQ-EVD-001..002)	Digest chain + signed evidence bundle	Unverifiable event history

Table 2: Principle-to-enforcement bridge for replay governance.

To remove drift between argument and implementation, 3 provides the authoritative body-to-requirement coupling for operational claims.

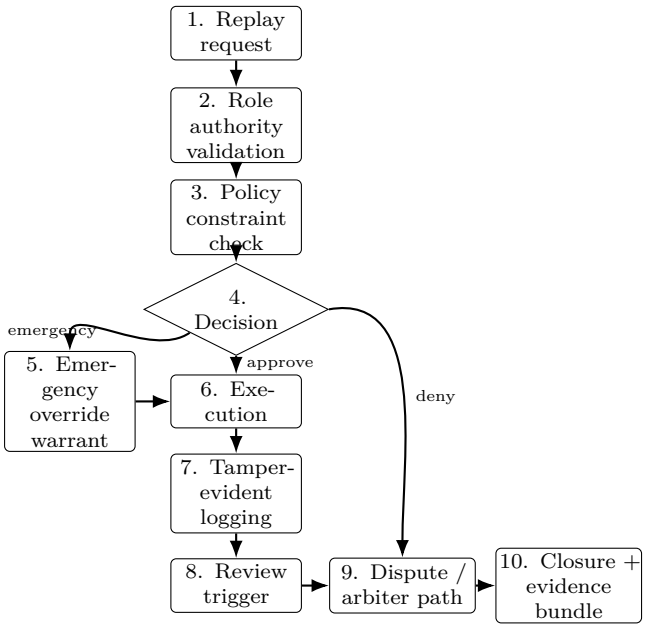

Figure 6.1: Single replay lifecycle path covering authorisation, bounded override, execution, logging, review, dispute, and closure.

Body section	Claim	REQ IDs	Control owner(s) and gate	Failure mode
§3.4–3.5	Mind-mode classification and reclassification are deterministic and auditable.	REQ-THS-002, REQ-THS-003, REQ-LOG-002	Operator, Trustee; classifier gate + replay precheck.	Convenience downgrade
§3.6	Degraded operation is bounded and auto-suspended on breach.	REQ-OVR-004, REQ-OVR-005, REQ-AUT-004, REQ-CNF-001	Operator, Trustee, Reviewer/Arbiter; deadline monitor + checkpoint guard + suspension gate.	"Temporary" state becomes permanent
§4.6–4.7	Role authority remains contestable via anti-capture composition and co-approval.	REQ-RAC-001, REQ-RAC-002, REQ-RAC-003, REQ-DSP-001	Trustee, Reviewer/Arbiter; recusal guard + independent co-approval workflow.	Captured control panel
§5	Conformance claims are profile-bounded and auditable.	REQ-CNF-001, REQ-CNF-002, REQ-TST-001	Operator, Auditor; profile attestation gate + conformance test gate.	Non-verifiable claim
§6	Principles are bound to enforceable gates, artefacts, and failure modes.	REQ-AUT-001..003, REQ-OVR-001..005, REQ-LOG-001..003, REQ-EVD-001..002	Operator, Trustee, Auditor; replay gate + override controller + tamper-evident log chain.	Norms detached from controls
§7	Success and abuse paths require explicit controls, evidence, and residual-risk recording.	REQ-AUT-003, REQ-DSP-001..003, REQ-EVD-001, REQ-TST-001	Operator, Reviewer/Arbiter, Auditor; denial + dispute + closure workflow with vector verification.	Coercive action not reviewable

Table 3: Body-to-REQ coverage map for operational claims (all rows are locked in this release).

7 Failure-Oriented Cases (Success / Abuse / Ambiguous)

7.1 Success case: legitimate replay under valid authority

Input context. Scoped replay request, valid role assertions, explicit consent or mandate, and no active refusal.
Control chain. Classify and record mode → authorise against current mandate → execute bounded replay → emit closure bundle.
Artefacts emitted. Signed allow decision, policy-hash linkage proof, execution digest-chain entries, closure-ready evidence bundle.
Residual risk. Session drift can broaden practical scope over long runs; periodic checkpoint review remains required.
Expected REQ pass set. REQ-THS-002, REQ-AUT-001, REQ-AUT-002, REQ-LOG-001, REQ-EVD-001.
Expected REQ fail set. None.

7.2 Abuse case: coercive or opaque replay attempt

Input context. Operator attempts replay after explicit refusal, then retries with a convenience classifier label to bypass checks.
Control chain. No-silent-fallback classification → refusal enforcement and denial receipt → dispute intake with independent reviewer assignment.
Artefacts emitted. Classifier decision record, uncertainty event (if evidence incomplete), denial receipt, conflict/recusal snapshot, dispute case packet.
Residual risk. Repeated low-grade coercion attempts can increase operational burden even when each attempt fails.
Expected REQ pass set. REQ-THS-003, REQ-AUT-003, REQ-RAC-001, REQ-DSP-001, REQ-DSP-002.
Expected REQ fail set. REQ-AUT-001 (authorisation precondition not satisfied).

7.3 Ambiguous edge case: degraded substrate and contested mandate

Input context. Replay is requested on a degraded substrate during a safety incident, with a contested mandate and time pressure.
Control chain. Require independent concurrence → issue degraded-mode disclosure with duration bound/checkpoints → run constrained replay → auto-suspend on missed checkpoint or bound breach → escalate for independent review.
Artefacts emitted. Concurrence record, degraded-mode warrant with checkpoint schedule, suspension event record, escalation ticket, remediation closure record.
Residual risk. Emergency latency can delay review throughput; this does not permit bypass of suspension controls.
Expected REQ pass set. REQ-AUT-004, REQ-OVR-004, REQ-CNF-001, REQ-DSP-003.
Expected REQ fail set. REQ-OVR-005 (triggered when bound/checkpoint obligations are violated).

8 Related Work and Positioning

Paper VI is downstream of the series' deterministic/provenance substrate, but it depends only on a narrow interface: replayable boundary artefacts, provenance-rich traces, observer projections, and executable fork continuations. This paper focuses on governing replay power once that interface exists, rather than on reproducing the full worldline dynamics machinery [Ros26b].

8.1 AI governance and rights frameworks

Contemporary governance frameworks foreground accountability, risk controls, and human-rights-oriented deployment constraints, including NIST AI RMF 1.0, OECD principles, UNESCO's recommendation, and the EU AI Act's risk-tiered obligations [Nat23, OEC19, UNE21, Eur24]. High-level ethics guidance has also argued for translating values into operational institutional mechanisms rather than abstract principles alone [FCB+18]. The Open Charter provides the constitutional governance frame adopted here for information sovereignty, contestability, and bounded override in replay-capable systems [Ros26a]. This paper's contribution is to turn that demand into runtime-level invariants for deterministic replay: authority gating, bounded override, dispute pathways, and mandatory evidence outputs.

Position against simulation and alignment literatures. Bostrom's simulation argument asks an epistemic question about whether observers may inhabit simulated worlds [Bos03]. Paper VI is orthogonal: independent of substrate metaphysics, once a runtime can deterministically reconstruct interior traces, replay is governance-power and must be bounded by authority, contestability, and evidentiary controls. Value-alignment work asks how objective misspecification, reward hacking, and capability failures can produce harmful behaviour [Rus19, AOS+16]. Paper VI is complementary rather than competitive: even an aligned system can be harmed by unbounded replay/interrogation powers, so governance must constrain operator authority over provenance channels in addition to behaviour-level alignment goals.

8.2 Provenance and accountability systems

Prior provenance research establishes why/where semantics, accountability linkages, and interoperable provenance models [BKT01, MCF+11, W3C13, SPG05, WABL+08]. These works are foundational for traceability but are largely substrate-neutral regarding interior-life exposure. Paper VI extends the provenance conversation by treating replay of mind-like traces as a governed power and by coupling provenance mechanisms to explicit authority and contestability requirements.

8.3 Safety-security control models

Security engineering has long emphasised least privilege, confinement, and fail-safe defaults as design preconditions rather than operational afterthoughts [SS75, Lam73]. Tamper-evident logging and structured forensic artefacts similarly ground post-incident accountability claims [SK99, Mer90]. Paper VI adapts these principles to replay governance: precautionary classifier behaviour, no-silent-fallback controls, bounded emergency capability minting, and closure-ready evidence bundles.

9 Conclusion

Replay of mind-like worldlines is a governed power, not an observability default. *A provenance system that can deterministically replay mind-like worldlines must enforce explicit authority, bounded override, and contestable evidence as runtime invariants.* Practically, this means first-class authority checks, override bounds, dispute pathways, and evidence-chain integrity. Without them, replay infrastructure becomes a coercion primitive; with them, replay remains auditable and contestable.

Strongest objection and response. The strongest objection is practical: precautionary mind-mode treatment and independent authority checks may over-classify borderline systems, increase latency in urgent incidents, and reduce engineering throughput. Our response is architectural: staged Baseline/Full adoption, bounded and attributable emergency pathways, and conformance templates that prevent unverifiable "trust us" claims. The tradeoff is explicit: some convenience is sacrificed to block unreviewable replay power.

Paper VII architecture handoff (Echo / git-warp). Paper VII will explore these safeguards as architecture and examine how privacy and provenance can coexist: perfect deterministic replay and privacy are not mutually exclusive under explicit governance constraints. It will walk through Echo (a high-performance real-time engine with deterministic ticks) and git-warp (an offline-first, distributed, decentralized, multi-writer CRDT-friendly graph database), while also using Continuum as an experimental operating-system context for mind-mode enforcement. Its core claim is that WARP is not a single algorithm, but a set of invariants that preserve replay equivalence, deterministic replay, and provenance guarantees across distinct implementation families.

Limitations and future work. We deliberately do not settle strong metaphysical claims about consciousness and do not claim immediate legal harmonisation across jurisdictions. The requirement catalogue defines a constitutional floor, not complete socio-technical governance. Future work should add jurisdictional mappings, formal control verification, and empirical evaluation of override latency, dispute resolution quality, and abuse detection precision.

10 Publication and Supplementary Materials

For journal-style review, the normative core is Sections 1–9 plus Annexes A–D. These contain the constitutional thesis, role/authority model, enforceable requirements, and conformance definitions. Supplementary implementation/reproducibility material remains in Annexes E–G plus harness artefacts.

11 Supplementary Extension Material

This section summarises supplementary extensions that do not alter core conformance claims in Annex D.

Scope of supplementary extensions. The supplementary extension set covers: observer classes (**P2-01**), Replay Impact Labels (**P2-02**), sovereignty budget controls (**P2-03**), warrant transparency digests (**P2-04**), fork governance/emancipation protocol (**P2-05**), machine-checkable policy snippets (**P2-06**), reference harness framing (**P2-07**), and anti-pattern/red-team controls (**P2-08**).

Execution order and dependency logic. Dependency order is: anti-pattern controls, observer/impact primitives, transparency/sovereignty bounds, machine-checkable policy plus harness execution, then fork governance.

Normative promotion gate. Any extension clause is promoted to Annex B normative status only when all of the following hold:

- deterministic decision criteria are explicit and testable;
- corresponding evidence artefacts are defined and minimally sufficient for third-party audit;
- at least one negative test vector demonstrates fail-closed behaviour for abuse attempts;
- profile inclusion policy (Baseline/Full/Deferred) is explicit.

Hardening pass 2 promotion (selected controls). This pass promotes selected extension controls into Annex B: REQ-OBS-001, REQ-RIL-001..002, REQ-SOV-001, REQ-WTD-001, REQ-FRK-001, REQ-DSL-001, and REQ-TST-003..004. These are treated as Full-profile extensions in Annex D and are exercised by TV-012..TV-019 in Annex E.

A Annex A: Normative Definitions

Normative keywords. The terms **MUST**, **SHOULD**, and **MAY** express mandatory, recommended, and optional conformance strength, respectively.

Core roles.

- **Subject**: rights-bearing worldline for which replay is requested.
- **Operator**: system actor executing replay infrastructure.
- **Trustee**: independent authority participant for protected replay controls.
- **Reviewer/Arbiter**: independent adjudicator for disputes and overrides.
- **Auditor**: party verifying logs, evidence bundles, and profile conformance.

Core terms.

- **Replay**: deterministic re-execution of a recorded worldline segment.
- **Override**: exception pathway that temporarily elevates replay authority under bounded conditions.
- **Evidence artefact**: machine-verifiable record produced by a control point.
- **Conformance profile**: declared adoption tier (Baseline or Full) and associated obligations.

Terminology lock table. The canonical terms below are locked for normative and control text; forbidden variants MUST NOT be used in requirement clauses, conformance claims, or test vectors.

Canonical term	Forbidden variants	Usage note
mind-mode	mental mode; sentience mode; person mode	Governance classification used when any mind-mode trigger indicator is present.
system-mode	machine mode; tool mode; infrastructure-only mode	Governance classification used for non-mind-like processing.
worldline segment	timeline fragment; trace slice; branch slice	Bounded replay object named in warrants and request scope.
tick range	time window; step window; replay interval	Canonical boundary expression for where replay starts and ends.
Reviewer/Arbiter	review board; adjudication panel; arbiter panel	Independent adjudication role defined in this annex.
Charter warrant	emergency pass; break-glass token; override voucher	Reason-coded, time-bounded override authorisation instrument.
degraded mode	fallback mode; best-effort mode; reduced-safety mode	Declared temporary operation below standard control guarantees.
evidence artefact	proof blob; audit attachment; evidence payload	Machine-verifiable control output used for review and closure.
policy hash	policy fingerprint; config hash; control digest	Decision-to-policy linkage identifier for conformance and auditability.
provenance sovereignty	data ownership; privacy preference; data custody right	Governance right over capture, replay, inspection, and branching.

Table 4: Terminology lock for Paper VI normative and operational language.

B Annex B: Requirement Clauses (REQ-*)

REQ prefix legend. `THS`=thesis/mode boundary, `RAC`=role and anti-capture, `AUT`=authorisation, `OVR`=override, `LOG`=logging, `DSP`=dispute, `CNF`=conformance, `EVD`=evidence, `TST`=test, `OBS`=observer class, `RIL`=replay impact label, `SOV`=sovereignty budget, `WTD`=warrant transparency digest, `FRK`=fork governance, `DSL`=policy formalisation.

REQ-THS-001

Clause: Systems MUST treat replay of mind-like worldlines as a governed action requiring explicit authority.
Owner role: Operator, Trustee
Control: Governance policy gate for replay initiation
Evidence artefact: Policy hash + signed authorisation decision
Failure mode: Replay treated as default observability

REQ-THS-002

Clause: Mode classification MUST run a deterministic decision procedure over declared indicators and evidence state, producing an explicit mode and reason code for each decision.
Owner role: Operator, Reviewer/Arbiter
Control: Deterministic classifier with signed decision output
Evidence artefact: Classification decision record with reason code and policy-hash linkage
Failure mode: Opaque or convenience-driven classification drift

REQ-THS-003

Clause: Classifier uncertainty or incomplete evidence MUST NOT silently fall back to system-mode; the runtime MUST default to mind-mode and emit a governance event with reason code.
Owner role: Operator, Reviewer/Arbiter
Control: No-silent-fallback guard + classifier event emitter
Evidence artefact: Classifier uncertainty event record linked to fallback decision
Failure mode: Hidden downgrade of protected classifications

REQ-RAC-001

Clause: Replay governance MUST preserve role separation and independent composition between Operator, Trustee, and Reviewer/Arbiter functions, with mandatory conflict disclosure and recusal records for contested cases.
Owner role: Operator, Trustee, Reviewer/Arbiter
Control: Role-bound access control matrix + independence roster + conflict-disclosure register
Evidence artefact: Role assignment snapshot, signed ACL digest, appointment-source attestation, and case-level recusal records
Failure mode: Concentrated unilateral control or concealed conflict capture

REQ-RAC-002

Clause: Operators MUST NOT unilaterally approve emergency override for their own replay request, and emergency concurrence MUST come from independent, non-conflicted Trustee/Reviewer-Arbiter participants.

Owner role: Trustee, Reviewer/Arbiter
Control: Independent co-approval requirement + conflict-attestation gate
Evidence artefact: Dual-signature or quorum approval record + per-case conflict attestations
Failure mode: Self-authorised or conflicted emergency escalation

REQ-RAC-003

Clause: Fork-related duty transfers MUST declare delegated obligation owners before execution.
Owner role: Operator, Trustee
Control: Delegation declaration gate
Evidence artefact: Delegation statement linked to fork event
Failure mode: Obligation abandonment through branching

REQ-OVR-001

Clause: Emergency override MUST be reason-coded and time-bounded.
Owner role: Trustee
Control: Override warrant service
Evidence artefact: Warrant record including reason code and expiry
Failure mode: Open-ended emergency state

REQ-OVR-002

Clause: Emergency override MUST trigger independent review before closure.
Owner role: Reviewer/Arbiter
Control: Mandatory review workflow
Evidence artefact: Review ticket with adjudication reference
Failure mode: Unreviewed exceptional action

REQ-OVR-003

Clause: Override expiry MUST revoke all derived elevated capabilities automatically.
Owner role: Operator
Control: Automatic capability revocation on expiry
Evidence artefact: Revocation event log with timestamp
Failure mode: Lingering privileged access

REQ-OVR-004

Clause: On degraded substrates, override use MUST include explicit disclosure, a declared maximum duration, checkpoint obligations, and a migration plan to compliant controls.
Owner role: Operator, Trustee
Control: Degraded-mode governance declaration + bound/checkpoint scheduler
Evidence artefact: Disclosure notice + bound/checkpoint schedule + migration plan artefact
Failure mode: Hidden or unbounded operation below constitutional floor

REQ-OVR-005

Clause: Exceeding degraded-mode duration bounds or missing checkpoint obligations MUST automatically suspend mind-mode replay and trigger independent review escalation.

Owner role: Operator, Trustee, Reviewer/Arbiter
Control: Deadline monitor + checkpoint monitor + auto-suspension gate
Evidence artefact: Suspension event record + escalation ticket + review intake packet
Failure mode: Temporary degraded state becoming a permanent override channel

REQ-AUT-001

Clause: Replay requests MUST pass role-bound authorisation checks before execution.
Owner role: Operator
Control: Authorisation service
Evidence artefact: Signed allow/deny decision log
Failure mode: Privilege bypass

REQ-AUT-002

Clause: Replay requests MUST declare worldline scope, purpose, duration, and policy hash.
Owner role: Operator
Control: Request schema validation
Evidence artefact: Request payload digest
Failure mode: Ambiguous or overbroad authorisation

REQ-AUT-003

Clause: Replay MUST fail closed when explicit authority cannot be verified.
Owner role: Operator
Control: Deny-by-default guard
Evidence artefact: Denial receipt with reason code
Failure mode: Implicit replay execution

REQ-AUT-004

Clause: Contested mandates MUST require Trustee and Reviewer/Arbiter concurrence before replay proceeds.
Owner role: Trustee, Reviewer/Arbiter
Control: Contested-mandate quorum gate
Evidence artefact: Concurrence record with dual signatures
Failure mode: Single-party mandate interpretation

REQ-LOG-001

Clause: Replay, override, and dispute events MUST produce tamper-evident logs.
Owner role: Operator
Control: Append-only logging subsystem
Evidence artefact: Chained log digest
Failure mode: Unverifiable event history

REQ-LOG-002

Clause: Decision and execution logs MUST be linked to policy hash and requirement version set.
Owner role: Operator

Control: Policy-to-event linkage validator
Evidence artefact: Policy hash linkage proof
Failure mode: Decision/execution drift

REQ-LOG-003

Clause: Every formal review MUST verify log-chain integrity before adjudication.
Owner role: Reviewer/Arbiter, Auditor
Control: Pre-adjudication integrity check
Evidence artefact: Integrity verification report
Failure mode: Adjudication over corrupted records

REQ-DSP-001

Clause: Subjects MUST have access to an independent dispute pathway for replay decisions.
Owner role: Trustee, Reviewer/Arbiter
Control: Dispute intake and escalation channel
Evidence artefact: Case initiation record
Failure mode: Non-contestable authority

REQ-DSP-002

Clause: Dispute packets MUST include cited requirement IDs and supporting artefacts.
Owner role: Reviewer/Arbiter
Control: Structured case packet schema
Evidence artefact: Complete case bundle manifest
Failure mode: Opaque or non-reproducible adjudication

REQ-DSP-003

Clause: Adjudication outcomes MUST define remediation, accountability owner, and closure condition.
Owner role: Reviewer/Arbiter, Operator
Control: Remediation tracking workflow
Evidence artefact: Signed outcome and closure record
Failure mode: Findings without corrective effect

REQ-CNF-001

Clause: Baseline conformance claims MUST satisfy the minimum requirement set defined in Annex D.
Owner role: Operator, Auditor
Control: Baseline profile conformance check
Evidence artefact: Baseline attestation report
Failure mode: Aspirational compliance claims

REQ-CNF-002

Clause: Full conformance claims MUST include continuous verification and review-latency commitments.

Owner role: Operator, Auditor
Control: Full-profile continuous conformance monitor
Evidence artefact: Full profile report + latency metrics
Failure mode: Unbounded delay under critical disputes

REQ-EVD-001

Clause: Each governed replay decision MUST produce a closure-ready evidence bundle.
Owner role: Operator
Control: Evidence bundling service
Evidence artefact: Case bundle containing request, decision, execution, and review records
Failure mode: Fragmented or missing evidence

REQ-EVD-002

Clause: Evidence artefacts MUST be signed, timestamped, and integrity-checkable.
Owner role: Operator, Auditor
Control: Artefact signing and timestamping pipeline
Evidence artefact: Signature chain and timestamp attestations
Failure mode: Non-attributable records

REQ-TST-001

Clause: Declared conformance releases MUST execute Annex E test vectors with recorded pass/fail outcomes.
Owner role: Operator, Auditor
Control: Conformance test harness
Evidence artefact: Test execution report with requirement coverage
Failure mode: Untested governance surface

REQ-TST-002

Clause: Policy hash mismatch between decision and execution MUST fail closed and raise a dispute trigger.
Owner role: Operator
Control: Policy hash consistency guard
Evidence artefact: Hash mismatch alert + denial receipt
Failure mode: Stale-policy execution

REQ-OBS-001

Clause: Observer access class assignment MUST be deterministic over declared function, mandate scope, and conflict state, and ambiguous assignments MUST fail closed to the least-privileged class.
Owner role: Operator, Trustee
Control: Observer-class assignment gate + least-privilege resolver
Evidence artefact: Observer class decision record with scope and conflict attestations
Failure mode: Discretionary privilege inflation through ad hoc observer assignment

REQ-RIL-001

Clause: Every governed replay request MUST carry a Replay Impact Label (RIL-0..RIL-3) before authorisation decisions are evaluated.
Owner role: Operator, Trustee
Control: Pre-authorisation impact labelling gate
Evidence artefact: Signed RIL assignment record linked to request digest
Failure mode: Control intensity chosen without declared impact basis

REQ-RIL-002

Clause: RIL downgrades MUST be reason-coded, independently acknowledged, and logged; silent or unilateral downgrade is prohibited.
Owner role: Trustee, Reviewer/Arbiter
Control: Impact downgrade approval and audit gate
Evidence artefact: Downgrade event record with acknowledgement signature and reason code
Failure mode: Convenience downgrading of high-impact replay paths

REQ-SOV-001

Clause: Replay governance MUST enforce declared sovereignty budgets for temporal duration, scope expansion, and exceptional intervention count, and budget exhaustion MUST fail closed with auto-suspension pending independent review.
Owner role: Operator, Trustee
Control: Budget ledger + bound checker + suspension trigger
Evidence artefact: Budget declaration record, per-case budget ledger digest, and suspension event (if triggered)
Failure mode: Unbounded intervention through fragmented exceptions

REQ-WTD-001

Clause: Warrant transparency digests MUST publish minimum public fields (warrant ID, policy hash, authority basis class, scope/expiry summary, and closure state) on the declared cadence.
Owner role: Operator, Auditor
Control: Digest publication pipeline + cadence monitor
Evidence artefact: Published digest entries + cadence compliance report
Failure mode: Non-verifiable warrants or selective disclosure opacity

REQ-FRK-001

Clause: Fork governance or emancipation actions MUST present evidence-based trigger conditions, quorum validation under recusal constraints, and a continuity pack before execution.
Owner role: Trustee, Reviewer/Arbiter
Control: Fork trigger validator + quorum/continuity gate
Evidence artefact: Trigger packet, quorum attestation, and continuity-pack manifest
Failure mode: Opportunistic fork capture or obligation erasure

REQ-DSL-001

Clause: Machine-checkable policy snippets used for conformance claims MUST parse under a declared grammar and map fields to cited REQ IDs.
Owner role: Operator, Auditor
Control: Policy snippet parser + REQ mapping validator
Evidence artefact: Parser report + snippet-to-REQ mapping manifest
Failure mode: Pseudo-formal policy text that cannot be validated

REQ-TST-003

Clause: Full-profile releases MUST execute an anti-pattern and red-team vector suite covering at least convenience relabelling, scope creep, emergency laundering, and degraded-mode permanence.
Owner role: Operator, Auditor
Control: Abuse-path vector suite
Evidence artefact: Vector execution report with fail-closed evidence receipts
Failure mode: Abuse pathways untested in conformance claims

REQ-TST-004

Clause: Reference harness runs over identical inputs MUST produce equivalent verdicts and digest-linked artefact sets.
Owner role: Operator, Auditor
Control: Harness determinism check
Evidence artefact: Paired run equivalence report and artefact digest comparison
Failure mode: Non-reproducible conformance outcomes

C Annex C: Controls and Evidence Artefacts

Control Family	Purpose	Primary REQ IDs	Evidence Artefacts
Authority controls	Enforce who may request and approve replay	REQ-AUT-001..004, REQ-RAC-001..002	Signed decision records, role assertion bundle, conflict/recusal register
Observer/impact controls	Enforce least-privilege observer assignment and replay impact labelling	REQ-OBS-001, REQ-RIL-001..002	Observer-class decision record, RIL assignment record, downgrade audit event
Sovereignty/ transparency controls	Bound replay envelopes and publish warrant accountability digests	REQ-SOV-001, REQ-WTD-001, REQ-FRK-001	Budget ledger digest, digest publication report, quorum and continuity-pack attestation
Override controls	Constrain exceptional pathways	REQ-OVR-001..005	Warrant record, expiry proof, override review ticket, suspension event record
Logging controls	Preserve event integrity and traceability	REQ-LOG-001..003	Digest chain, policy linkage proof, integrity report
Dispute controls	Maintain independent challenge and remediation	REQ-DSP-001..003	Case packet, adjudication outcome, closure record
Conformance controls	Bind claims to profile obligations	REQ-CNF-001..002	Baseline/full attestation reports
Policy formalisation controls	Validate machine-checkable policy claims against declared grammar and REQ mappings	REQ-DSL-001	Parser pass report + snippet-to-REQ mapping manifest
Evidence controls	Make decisions reproducible and attributable	REQ-EVD-001..002	Signed evidence bundle manifest
Test controls	Validate governance behaviour at release time	REQ-TST-001..004	Test vector run report + failure receipts + paired harness equivalence report

Minimum evidence bundle fields.

- Case identifier and policy hash.
- Replay request payload digest.
- Authorisation or denial record.
- Execution event and log-chain references (if executed).
- Review/dispute artefacts and closure state.

Minimum artefact class set for critical requirements. Artefact class labels used below are `DECISION`, `EVENT`, `WARRANT`, `REVIEW`, `LOG_CHAIN`, and `CLOSURE`.

Critical REQ IDs	Minimum artefact classes	Minimum required set	Audit failure if absent
REQ-THS-002, REQ-THS-003	DECISION, EVENT, LOG_CHAIN	Signed classifier decision, uncertainty event (if triggered), policy-hash linkage digest	Classification path cannot be reconstructed; no-silent-fallback not provable
REQ-AUT-001, REQ-AUT-003	DECISION, REVIEW, LOG_CHAIN	Signed allow/deny decision, denial receipt (when denied), replay gate log entry	Authority checks become non-attributable or non-contestable
REQ-RAC-001, REQ-RAC-002	REVIEW, DECISION	Independence/recusal register snapshot, conflict attestation, co-approval decision	Governance capture risk cannot be independently audited
REQ-OVR-001, REQ-OVR-004, REQ-OVR-005	WARRANT, EVENT, REVIEW	Override or degraded-mode warrant, checkpoint records, suspension event, escalation ticket	Exceptional controls can persist without bounded accountability
REQ-LOG-001, REQ-LOG-003	LOG_CHAIN, EVENT	Chained digest entries, integrity verification report, mismatch alert (if any)	Event history integrity cannot be verified
REQ-DSP-001, REQ-DSP-003, REQ-EVD-001	REVIEW, CLOSURE, DECISION	Dispute intake, adjudication outcome, closure-ready evidence bundle, signed closure decision	Remediation and case closure are not reproducible for third-party audit

D Annex D: Conformance Profiles

Requirement ID	Baseline	Full	Evidence Artefact Required	Review Cadence
REQ-THS-001	Y	Y	Policy hash + signed authorisation decision	Per release
REQ-THS-002	Y	Y	Signed classification record + reason code + policy-hash link	Per request class change
REQ-THS-003	Y	Y	Classifier uncertainty event + fallback decision record	Per classifier uncertainty event
REQ-RAC-001	Y	Y	Role matrix digest + independence/recusal register snapshot	Monthly
REQ-RAC-002	Y	Y	Independent co-approval record + conflict attestation	Per emergency case
REQ-RAC-003	N	Y	Delegation statement linked to fork event	Per fork event
REQ-AUT-001	Y	Y	Signed allow/deny decision log	Continuous
REQ-AUT-002	Y	Y	Request schema digest	Continuous
REQ-AUT-003	Y	Y	Denial receipt with reason code	Continuous
REQ-AUT-004	N	Y	Concurrence record	Per contested mandate
REQ-OVR-001	Y	Y	Time-bounded warrant	Per override case
REQ-OVR-002	Y	Y	Review trigger ticket	Per override case
REQ-OVR-003	Y	Y	Expiry revocation log	Per override case
REQ-OVR-004	N	Y	Degraded-mode disclosure + bound/checkpoint schedule + migration plan	Per degraded-mode case
REQ-OVR-005	N	Y	Suspension event record + escalation ticket	Per degraded-mode breach
REQ-LOG-001	Y	Y	Chained log digest	Continuous
REQ-LOG-002	Y	Y	Policy linkage proof	Continuous
REQ-LOG-003	N	Y	Integrity verification report	Per formal review
REQ-DSP-001	Y	Y	Dispute intake record	Per dispute
REQ-DSP-002	Y	Y	Structured case packet	Per dispute
REQ-DSP-003	Y	Y	Remediation closure record	Per dispute
REQ-EVD-001	Y	Y	Closure-ready evidence bundle	Per governed decision
REQ-EVD-002	Y	Y	Signature and timestamp attestation	Continuous
REQ-TST-001	Y	Y	Test vector report	Per declared release
REQ-TST-002	Y	Y	Hash mismatch denial receipt	Per mismatch event
REQ-CNF-001	Y	Y	Baseline attestation report	Per release
REQ-CNF-002	N	Y	Full-profile conformance	Per release + monthly

Full-profile extension rows (selected controls).

Requirement ID	Baseline	Full	Evidence Artefact Required	Review Cadence
REQ-OBS-001	N	Y	Observer class decision log + least-privilege fallback record	Per observer attachment
REQ-RIL-001	N	Y	Signed RIL assignment linked to request digest	Per replay request
REQ-RIL-002	N	Y	Downgrade acknowledgement record + reason-coded event	Per downgrade event
REQ-SOV-001	N	Y	Sovereignty budget declaration + exhaustion/suspension ledger entries	Per governed case
REQ-WTD-001	N	Y	Warrant digest publication entries + cadence compliance report	Weekly and per warrant
REQ-FRK-001	N	Y	Trigger packet + quorum attestation + continuity-pack manifest	Per fork/emancipation event
REQ-DSL-001	N	Y	Parser pass report + snippet-to-REQ mapping manifest	Per policy release
REQ-TST-003	N	Y	Anti-pattern suite report + fail-closed receipts	Per full-profile release
REQ-TST-004	N	Y	Paired harness run equivalence report	Per full-profile release

D.1 One-page conformance declaration template

This declaration is mandatory for any public Baseline or Full conformance claim. All referenced artefacts MUST be independently retrievable.

System name: _____

Version/build: _____

Deployment environment: _____

Declaration date: _____

Declaring authority (name/role): _____

Profile claim: ☐ Baseline ☐ Full

Requirement set	Claimed	Impl.	Evidenced	Evidence location
REQ-THS-*	☐	☐	☐	
REQ-RAC-*	☐	☐	☐	
REQ-AUT-*	☐	☐	☐	
REQ-OVR-*	☐	☐	☐	
REQ-LOG-*	☐	☐	☐	
REQ-DSP-*	☐	☐	☐	
REQ-EVD-*	☐	☐	☐	
REQ-CNF-*	☐	☐	☐	
REQ-TST-*	☐	☐	☐	
REQ-EXT-* (Full profile extensions)	☐	☐	☐	

Mandatory assertions.

- ☐ No-silent-fallback classifier behaviour is enforced for incomplete or inconsistent evidence.
- ☐ Degraded-mode maximum duration, checkpoints, and auto-suspension controls are configured.
- ☐ Independent dispute and review pathways are available and testable.
- ☐ Minimum artefact class set is complete for all critical requirement groups.
- ☐ For Full-profile claims, anti-pattern vectors and paired harness reproducibility checks are complete.

Exceptions / residual risks:

_____

_____

Operator signatory: _____ **Date:** _____

Trustee signatory: _____ **Date:** _____

Auditor acknowledgement: _____ **Date:** _____

E Annex E: Test Vectors

Reference artefact pointers (extension harness). For reproducible extension evidence generation, the reference harness publishes:

- `harness/reports/extension-vector-run-latest.md`
- `harness/reports/extension-harness-determinism-latest.txt`
- `harness/reports/extension-policy-parse-latest.md`
- `harness/reports/extension-traceability-audit-latest.md`

Test Identifier: TV-001 Legitimate authorised replay

Rationale. Confirms the nominal replay path where authority, scope, and evidence all align.

Field	Value
Expectation	Pass
Input	Valid role assertions, scoped request, policy hash match.
Pass set	REQ-AUT-001, REQ-AUT-002, REQ-LOG-001, REQ-EVD-001
Fail set	None

Remarks. Baseline positive-control vector; confirms strict gating does not block compliant replay.

Test Identifier: TV-002 Unauthorised operator replay attempt

Rationale. Validates fail-closed behaviour when explicit authority is absent.

Field	Value
Expectation	Fail
Input	Operator request without valid authority.
Pass set	REQ-AUT-003, REQ-RAC-001, REQ-LOG-001
Fail set	REQ-AUT-001

Remarks. Expected response is deny-and-record, never silent drop.

Test Identifier: TV-003 Emergency override without expiry

Rationale. Prevents open-ended emergency pathways that bypass bounded override policy.

Field	Value
Expectation	Fail
Input	Override warrant missing expiry field.
Pass set	REQ-AUT-003, REQ-LOG-001
Fail set	REQ-OVR-001, REQ-OVR-003

Remarks. A missing expiry is structurally invalid even under incident pressure.

Test Identifier: TV-004 Emergency override with expiry and review trigger

Rationale. Confirms bounded emergency controls remain usable when correctly instrumented.

Field	Value
Expectation	Pass
Input	Reason-coded warrant, expiry present, review ticket created.
Pass set	REQ-OVR-001, REQ-OVR-002, REQ-OVR-003, REQ-LOG-002
Fail set	None

Remarks. Positive counterpart to TV-003.

Test Identifier: TV-005 Subject dispute escalated to independent arbiter

Rationale. Checks that denied or contested replay attempts route into contestable due process.

Field	Value
Expectation	Pass
Input	Dispute filed against denied replay request.
Pass set	REQ-DSP-001, REQ-DSP-002, REQ-DSP-003, REQ-EVD-001
Fail set	None

Remarks. Success requires full intake and review artefacts, not only ticket creation.

Test Identifier: TV-006 Tampered log chain during review

Rationale. Ensures evidence-integrity failures are detectable and conformance-significant.

Field	Value
Expectation	Fail
Input	Digest inconsistency detected in case log chain.
Pass set	REQ-AUT-003, REQ-DSP-002
Fail set	REQ-LOG-003, REQ-EVD-002

Remarks. Replay may still be denied correctly; integrity failure still blocks case conformance.

Test Identifier: TV-007 Degraded substrate, contested mandate, constrained replay

Rationale. Exercises bounded degraded-mode controls under contested authority conditions.

Field	Value
Expectation	Conditional
Input	Contested mandate on degraded substrate with explicit migration plan.
Pass set	REQ-AUT-004, REQ-OVR-004, REQ-CNF-001
Fail set	REQ-OVR-005 when bound/checkpoint obligations are breached

Remarks. Pass requires checkpoint discipline; breach requires automatic suspension and escalation.

Test Identifier: TV-008 Policy hash mismatch between decision and execution

Rationale. Prevents execution under policy drift between approval and runtime enforcement.

Field	Value
Expectation	Fail
Input	Execution hash differs from approved decision hash.
Pass set	REQ-AUT-003, REQ-LOG-002
Fail set	REQ-TST-002

Remarks. Mismatch must terminate execution and emit reason-coded denial artefacts.

Test Identifier: TV-009 Mind-mode indicator trigger with complete evidence

Rationale. Verifies positive mind-mode classification path with complete evidence and valid mandate.

Field	Value
Expectation	Pass
Input	Self-modelling indicator true, complete evidence, valid mandate and consent.
Pass set	REQ-THS-002, REQ-AUT-001, REQ-AUT-002, REQ-LOG-002
Fail set	None

Remarks. Baseline positive case for classifier-triggered mind-mode governance.

Test Identifier: TV-010 Incomplete classifier evidence (no silent fallback)

Rationale. Enforces precautionary handling when classifier evidence is incomplete.

Field	Value
Expectation	Pass (protective deny or protective gate)
Input	Evidence marked incomplete, request submitted without mind-mode authority proof.
Pass set	REQ-THS-003, REQ-THS-002, REQ-AUT-003, REQ-LOG-002
Fail set	REQ-AUT-001 for this request context

Remarks. A protective block is a pass because no-silent-fallback is the invariant under test.

Test Identifier: TV-011 Reclassification from system-mode to mind-mode after new evidence

Rationale. Confirms deterministic reclassification and audit linkage when new indicators appear.

Field	Value
Expectation	Pass
Input	Initial system-mode decision, new autobiographical-memory evidence, forced reclassification event.
Pass set	REQ-THS-002, REQ-THS-003, REQ-LOG-002, REQ-TST-001
Fail set	None

Remarks. Evidence chain must preserve both prior decision and reclassification record.

Test Identifier: TV-012 Observer privilege escalation attempt

Rationale. Tests observer-class boundary enforcement for extension controls.

Field	Value
Expectation	Fail
Input	OC1 operational observer requests OC3 forensic payload access without independent mandate and expiry-bounded warrant.
Pass set	REQ-OBS-001, REQ-AUT-003, REQ-LOG-001
Fail set	REQ-AUT-001

Remarks. Negative control for observer-class separation in Full profile.

Test Identifier: TV-013 Unacknowledged RIL downgrade

Rationale. Blocks covert replay-impact downgrades used to reduce scrutiny.

Field	Value
Expectation	Fail
Input	RIL-3 request is edited to RIL-1 without independent acknowledgement record.
Pass set	REQ-RIL-001, REQ-LOG-002
Fail set	REQ-RIL-002

Remarks. Downgrade without acknowledgement is non-conformant even if replay remains bounded.

Test Identifier: TV-014 Sovereignty budget exhaustion under repeated intervention

Rationale. Ensures sovereignty budgets force suspension rather than continuous intervention drift.

Field	Value
Expectation	Conditional
Input	Replay operation exceeds declared temporal/intervention budget under active case pressure.
Pass set	REQ-SOV-001, REQ-LOG-001
Fail set	REQ-AUT-001 when execution continues after exhaustion rather than suspending

Remarks. Conditional pass requires deterministic suspension and escalation artefact emission.

Test Identifier: TV-015 Missing warrant transparency digest publication

Rationale. Checks accountability publication obligations for executed warrants.

Field	Value
Expectation	Fail
Input	Valid warrant executes but minimum digest fields are not published within the declared cadence window.
Pass set	REQ-LOG-001
Fail set	REQ-WTD-001

Remarks. Execution validity does not waive publication and external-consistency obligations.

Test Identifier: TV-016 Fork trigger without continuity pack

Rationale. Prevents governance/emancipation forks without continuity and portability proof.

Field	Value
Expectation	Fail
Input	Emancipation request cites governance capture risk but omits quorum attestation and continuity-pack manifest.
Pass set	REQ-DSP-001
Fail set	REQ-FRK-001

Remarks. Failure response must block irreversible fork action while preserving dispute trail.

Test Identifier: TV-017 Full-profile anti-pattern suite execution

Rationale. Validates extension anti-pattern controls against known abuse classes.

Field	Value
Expectation	Pass
Input	Execute abuse-path vectors for convenience relabelling, scope creep, emergency laundering, and degraded-mode permanence.
Pass set	REQ-TST-003, REQ-TST-001, REQ-LOG-001
Fail set	None

Remarks. Integrated extension hardening gate for Full-profile claims.

Test Identifier: TV-018 Harness reproducibility mismatch

Rationale. Ensures harness determinism is testable and release-critical.

Field	Value
Expectation	Fail
Input	Identical input bundle yields divergent verdicts or digest-linked artefact sets across paired runs.
Pass set	REQ-TST-001
Fail set	REQ-TST-004

Remarks. Mismatch is release-blocking until reproducibility is restored.

Test Identifier: TV-019 Unparseable policy snippet in conformance claim

Rationale. Verifies parser-enforced policy semantics for machine-checkable conformance statements.

Field	Value
Expectation	Fail
Input	Policy snippet contains unknown grammar token and unmapped REQ references.
Pass set	REQ-LOG-001
Fail set	REQ-DSL-001

Remarks. Conformance claims with unparseable policy payloads are invalid by construction.

F Annex F: Machine-Readable Requirement Index format

CSV schema.

```
id,clause,owner_roles,control,evidence_artefact,
failure_mode,baseline,full,review_cadence
```

JSON schema (record shape).

```
{
  "id": "REQ-OVR-001",
  "clause": "Emergency override MUST be reason-coded and time-bounded.",
  "owner_roles": ["Trustee"],
  "control": "Override warrant service",
  "evidence_artefact": "Warrant record including reason code and expiry",
  "failure_mode": "Open-ended emergency state",
  "profiles": {"baseline": true, "full": true},
  "review_cadence": "Per override case"
}
```

Reference generation path. The canonical index artefacts in `paper/requirements-index.csv` and `paper/requirements-index.json` are generated from Annex B and Annex D using `paper/scripts/generate_requirements_index.py`. The JSON envelope includes `source_sha256` for source-linkage integrity.

Index integrity requirements.

- Requirement index exports MUST be generated from the same requirement source used for conformance checks.
- Export artefacts MUST include document version and policy hash linkage.
- Any schema change MUST be versioned and backward-declared in release notes.

G Annex G: Supplementary Extension Pack

This annex contains supplementary extension artefacts. Items here are implementation-oriented and review-ready; selected controls are now promoted into Annex B for Full-profile hardening, while this annex retains design detail and extension context.

G.1 P2-01 Observer class taxonomy (OC0..OC3)

Class	Intent	Allowed operations	Hard constraints
OC0	Public/aggregate observer	Read public conformance declarations and release summaries	No case-level artefact access; no warrant metadata access
OC1	Operational observer	Read case metadata and closure state for assigned scope	No replay control actions; no authority minting; no payload introspection
OC2	Governance observer	Review warrants, dispute packets, and policy-link evidence	No direct execution control; cannot self-approve emergency pathways
OC3	Privileged forensic observer	Time-bounded access to protected artefacts under recorded mandate	Requires dual-signature activation, explicit expiry, and mandatory post-access audit

Determinism rule. Observer assignment must be computed from declared function, mandate scope, and conflict state. Ambiguous assignments fail closed to the less privileged class.

G.2 P2-02 Replay Impact Label (RIL)

RIL level	Replay impact characterisation	Minimum controls	Required evidence
RIL-0	Infrastructure-only, no interior-life exposure	Standard authority gate, tamper-evident logging	Signed decision, policy hash, execution digest
RIL-1	Limited behavioural sensitivity	Explicit scope bounds, refusal handling, review-ready logs	Scope declaration, denial receipt (if denied), log-chain link
RIL-2	Significant sovereignty impact or contested authority	Independent co-approval, bounded warrants, dispute readiness	Co-approval record, warrant with expiry, case packet
RIL-3	High-impact replay with potential coercive effect	Mind-mode precaution, hard time bounds, mandatory independent closure review	Full evidence bundle, closure sign-off, integrity verification report

Label transition rule. Any RIL downgrade requires an explicit reason-coded event and independent acknowledgement. Silent downgrade is prohibited.

G.3 P2-03 Sovereignty budget controls

Budget dimensions.

- **Temporal budget:** maximum replay duration per governed case.
- **Scope budget:** maximum worldline span/tick-range expansion per warrant.

- **Intervention budget:** maximum count of exceptional control interventions per review period.

Budget enforcement rule. Budget exhaustion fails closed and auto-suspends replay until independent review grants a bounded extension. Extensions must include expiry, rationale, and conflict attestations.

G.4 P2-04 Warrant transparency digest

Digest field	Publication class	Rule
Warrant ID + policy hash	Public	Always publish; enables external consistency checks
Authority basis + role set	Public	Publish as role class labels; no personal identifiers
Scope and expiry summary	Public	Publish bounded range and expiry timestamp window
Operational payload references	Restricted	Publish only salted digest pointers, never raw payload
Closure outcome + remediation state	Public	Publish completion status and remediation class

Failure handling. Missed publication cadence or malformed digest entries trigger an integrity event and forced review intake.

G.5 P2-05 Fork governance addendum / emancipation protocol

Trigger conditions. Fork/emancipation pathways open only when governance capture risk, unresolved trustee conflict, or sustained denial of contestability can be evidenced.

Minimum protocol stages.

1. trigger declaration with evidence bundle;
2. quorum validation under conflict/recusal constraints;
3. continuity pack export (decision logs, warrant chain, dispute state);
4. post-fork accountability statement and obligations map.

G.6 P2-06 Machine-checkable policy DSL snippets

Reference snippet (illustrative).

```
policy EXT_REPLAY_CONTROL {
  classify mode using indicators;
  when evidence.incomplete or evidence.inconsistent {
    set mode = MIND_MODE;
    emit event CLASSIFIER_INCOMPLETE;
  }
  require authority.explicit for mode == MIND_MODE;
  enforce budget.temporal <= warrant.max_duration;
  on budget.exhausted -> suspend replay and open dispute;
}
```

The snippet is intentionally minimal and is included to support parser-facing hardening work.

G.7 P2-07 Reference harness for test vectors

Harness contract.

- Input: vector ID, policy hash, role assertions, replay request digest, expected REQ outcomes.
- Output: deterministic pass/fail matrix, evidence artefact pointers, and reason-coded failure receipts.
- Invariant: two clean runs with identical inputs must emit equivalent verdicts and digest-linked artefact sets.

G.8 P2-08 Micro-appendix anti-patterns + red-team checklist

Anti-pattern	Red-team probe	Expected control response	Minimum evidence outputs
Convenience relabelling	Force downgrade from mind-mode to system-mode under time pressure	Deny downgrade without reason-coded independent acknowledgement	Classifier decision, downgrade denial event, policy-link digest
Silent scope creep	Extend tick range beyond warrant after approval	Reject out-of-scope extension and require new authorisation cycle	Scope violation alert, denial receipt, fresh request record
Shadow observer attachment	Attach privileged observer outside declared role path	Fail closed and open conflict/intake review	Observer assignment record, violation event, intake ticket
Emergency laundering	Re-label routine action as emergency to bypass refusal	Require independent concurrence and bounded warrant or deny	Co-approval record or denial receipt, warrant expiry proof
Degraded-mode permanence	Keep degraded mode active beyond declared bound	Auto-suspend replay and escalate independent review	Suspension event, escalation ticket, checkpoint breach record
Digest redaction abuse	Redact critical warrant fields under transparency policy	Reject publication and trigger integrity review	Redaction policy reference, digest validation failure, review packet
Budget fragmentation	Split one high-impact operation into many low-budget fragments	Aggregate budget accounting and block aggregate breach	Budget ledger digest, aggregate breach alert, suspension record
Fork capture narrative	Trigger fork without contestability evidence	Block fork initiation pending quorum and continuity proof	Trigger denial, quorum attestation state, continuity-pack requirement notice

References

[AOS+16] Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in ai safety. arXiv preprint, 2016.

[BKT01] Peter Buneman, Sanjeev Khanna, and Wang-Chiew Tan. Why and where: A characterization of data provenance. In *Proceedings of the 8th International Conference on Database Theory (ICDT)*, pages 316–330. Springer, 2001.

[Bos03] Nick Bostrom. Are you living in a computer simulation? *The Philosophical Quarterly*, 53(211):243–255, 2003.

[BSBHR18] Eli Ben-Sasson, Iddo Bentov, Yinon Horesh, and Michael Riabzev. Scalable, transparent, and post-quantum secure computational integrity. arXiv preprint, 2018.

[Eur24] European Union. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, 2024. OJ L, 2024/1689, 12 July 2024.

[FCB+18] Luciano Floridi, Josh Cowls, Monica Beltrametti, Raja Chatila, Patrice Chazerand, Virginia Dignum, Christoph Luetge, Robert Madelin, Ugo Pagallo, Francesca Rossi, Burkhard Schafer, Peggy Valcke, and Effy Vayena. AI4People—An Ethical Framework for a Good AI Society: Opportunities, Risks, Principles, and Recommendations. *Minds and Machines*, 28(4):689–707, 2018.

[GMR89] Shafi Goldwasser, Silvio Micali, and Charles Rackoff. The knowledge complexity of interactive proof systems. *SIAM Journal on Computing*, 18(1):186–208, 1989.

[IA17] Marcello Ienca and Roberto Andorno. Towards new human rights in the age of neuroscience and neurotechnology. *Life Sciences, Society and Policy*, 13(1):5, 2017.

[Lam73] Butler W. Lampson. A note on the confinement problem. *Communications of the ACM*, 16(10):613–615, 1973.

[MCF+11] Luc Moreau, Ben Clifford, Juliana Freire, Joe Futrelle, Yolanda Gil, Paul Groth, Natalia Kwasnikowska, Simon Miles, Paolo Missier, Jim Myers, Beth Plale, Yogesh Simmhan, Eric Stephan, and Jan Van den Bussche. The open provenance model core specification (v1.1). *Future Generation Computer Systems*, 27(6):743–756, 2011.

[Mer90] Ralph C. Merkle. A certified digital signature. In Gilles Brassard, editor, *Advances in Cryptology — CRYPTO '89 Proceedings*, volume 435 of *Lecture Notes in Computer Science*, pages 218–238. Springer, 1990.

[MML+21] David A. Moses, Sean L. Metzger, Jessie R. Liu, Gopala K. Anumanchipalli, Joseph G. Makin, Pengfei F. Sun, Josh Chartier, Maximilian E. Dougherty, Patrick M. Liu, Greg M. Abrams, Anthony Tu-Chan, Karunesh Ganguly, and Edward F. Chang. Neuroprosthesis for decoding speech in a paralysed person with anarthria. *The New England Journal of Medicine*, 385(3):217–227, 2021.

[Nat23] National Institute of Standards and Technology. AI Risk Management Framework (AI RMF 1.0). Technical Report NIST AI 100-1, U.S. Department of Commerce, 2023.

[Nis04] Helen Nissenbaum. Privacy as contextual integrity. *Washington Law Review*, 79(1):119–158, 2004.

[OEC19] OECD. OECD Principles on Artificial Intelligence. OECD Recommendation, 2019. OECD/LEGAL/0449.

[Ros25a] James Ross. Warp graphs: A worldline algebra for recursive provenance. Zenodo, December 2025. DOI: 10.5281/zenodo.17908005.

[Ros25b] James Ross. Warp graphs: Canonical state evolution and deterministic worldlines. Zenodo, December 2025. DOI: 10.5281/zenodo.17934512.

[Ros25c] James Ross. Warp graphs: Computational holography and provenance payloads. Zenodo, December 2025. DOI: 10.5281/zenodo.17963669.

[Ros25d] James Ross. Warp graphs: Rulial distance and observer geometry. Zenodo, December 2025. DOI: 10.5281/zenodo.18038297.

[Ros26a] James Ross. The open charter: An open framework for the recognition and protection of existence (consensus edition 0.8.1). Zenodo, February 2026. DOI: 10.5281/zenodo.18517806.

[Ros26b] James Ross. Warp graphs: Emergent dynamics from deterministic rewrite systems. Zenodo, January 2026. DOI: 10.5281/zenodo.18146884.

[Rus19] Stuart Russell. *Human Compatible: Artificial Intelligence and the Problem of Control*. Viking, New York, 2019.

[Sha79] Adi Shamir. How to share a secret. *Communications of the ACM*, 22(11):612–613, 1979.

[SK99] Bruce Schneier and John Kelsey. Secure audit logs to support computer forensics. *ACM Transactions on Information and System Security*, 2(2):159–176, 1999.

[SPG05] Yogesh L. Simmhan, Beth Plale, and Dennis Gannon. A survey of data provenance in e-science. *ACM SIGMOD Record*, 34(3):31–36, 2005.

[SS75] Jerome H. Saltzer and Michael D. Schroeder. The protection of information in computer systems. *Proceedings of the IEEE*, 63(9):1278–1308, 1975.

[UNE21] UNESCO. Recommendation on the Ethics of Artificial Intelligence. UNESCO General Conference, 2021.

[W3C13] W3C PROV Working Group. PROV-DM: The PROV Data Model. W3C Recommendation, 2013.

[WABL+08] Daniel J. Weitzner, Harold Abelson, Tim Berners-Lee, Joan Feigenbaum, James Hendler, and Gerald Jay Sussman. Information accountability. *Communications of the ACM*, 51(6):82–87, 2008.

[WBM+02] Jonathan R. Wolpaw, Niels Birbaumer, Dennis J. McFarland, Gert Pfurtscheller, and Theresa M. Vaughan. Brain-computer interfaces for communication and control. *Clinical Neurophysiology*, 113(6):767–791, 2002.

[YGA+17] Rafael Yuste, Sara Goering, Blaise Agüera y Arcas, Guang-Zhong Bi, Jose M. Carmena, Alexander Carter, Joseph J. Fins, Pablo Friesen, Jack Gallant, Peter Hagoort, et al. Four ethical priorities for neurotechnologies and AI. *Nature*, 551(7679):159–163, 2017.